MODEL 621

LASER WAVELENGTH METER

Version 4.0

USER'S MANUAL

The information contained in this publication is subject to change without notice.

CONTENTS

1 INTRODUCTION

Welcome

Thank you for purchasing the 621 Laser Wavelength Meter from Bristol Instruments. Two different models of this system are available, the 621A and the 621B. The model 621A measures the absolute wavelength of CW lasers to an accuracy of ± 0.2 parts per million (± 0.0002 nm at 1000 nm). The 621B system has an accuracy of ± 0.75 parts per million (± 0.00075 nm at 1000 nm). Both systems are available for operation over the wavelength range of 350 – 1100 nm (VIS version), 500 – 1700 nm (NIR version), or 1.0 – 5.0 μm (IR version). The 621B system is also available for operation over the wavelength range of 4 – 11 μm (MIR version) with an accuracy of ± 1.0 part per million (± 0.006 nm at 6 μm).

This *User's Manual* includes information about the 621A and 621B Laser Wavelength Meters. It covers all the topics necessary to help you operate your system.

If you have any questions about the operation of your 621 system, please do not hesitate to call Bristol Instruments at (585) 924-2620. Or, you can contact us at service@bristol-inst.com.

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Safety Notices

WARNING

Warning denotes a hazard. It calls attention to a procedure that, if not correctly performed or adhered to, could result in injury or loss of life. Do not proceed beyond a warning sign until the indicated conditions are fully understood and met.

CAUTION

Caution denotes a hazard. It calls attention to a procedure that, if not correctly performed or adhered to, could result in damage to or destruction of the product. Do not proceed beyond a caution sign until the indicated conditions are fully understood and met.

General Safety Considerations

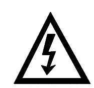
This product has been designed and tested in accordance with IEC 61010-1 and has been supplied in a safe condition. The instruction documentation contains information and warnings that must be followed by the user to ensure safe operation and to maintain the product in a safe condition.

WARNING **This instrument is a Class 2 Laser Product. It contains a laser that emits radiation that can seriously damage your eyesight. Less than 15 µW of this laser light is emitted from the fiber-optic connector of the VIS and NIR versions, and less than 50 µW of this laser light is emitted from the aperture of the IR and MIR versions. Do not stare into this beam or view it directly with an optical instrument. The class 2 warning label, shown here, is located next to the fiber-optic input coupler (VIS and NIR) or the input aperture (IR and MIR) on the front panel of the model 621.**

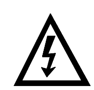
WARNING ***Never* inspect or clean a fiber-optic cable without first disconnecting the entire cable assembly from the optical source. Failure to take this precaution can seriously damage your eyesight.**

WARNING **If this instrument is not used as specified, the protection provided by the equipment could be impaired. This instrument must be used in the specified mode without deviation from the written instructions.**

WARNING **No user serviceable parts are inside. Refer servicing to Bristol Instruments or its representatives only. To prevent electrical shock, do not remove covers.**



WARNING **To prevent electrical shock, disconnect this instrument from mains before cleaning. Use a dry cloth or one slightly dampened with water to clean the external case parts. Do not attempt to clean internally.**



WARNING **This is a Safety Class 1 product (provided with protective earth). The mains plug shall only be inserted in a socket outlet provided with a protective earth contact. Any interruption of the protective conductor inside or outside of the product is likely to make the product dangerous. Intentional interruption is prohibited.**

CAUTION Always use the three-prong AC power cord supplied with this instrument. Failure to ensure adequate earth grounding by not using this cord may cause instrument damage.

CAUTION *Do not* connect AC power until you have verified the line voltage is within the specified range; 90 – 260 VAC, 47 – 63 Hz, protective earth ground. Damage to the instrument could result.

CAUTION This instrument has autoranging line voltage input. Be sure the supply voltage is within the specified range.

CAUTION Maximum safe input is 10 mW of optical power. Laser input power in excess of 10 mW can result in damage to the instrument.

CAUTION Use care in handling fiber-optic connectors. Always clean the fiber end prior to insertion into the instrument's fiber-optic connector for optimum performance. Failure to do so can result in damage to the instrument. To prolong instrument service life, it is advantageous to attach a new fiber-optic patchcord to the instrument and use its other end for the connect/disconnect procedure.

Unpacking and Inspection

The 621 Laser Wavelength Meter is packed in a carton designed to give maximum protection during shipment. If the outside of the shipping carton is damaged, notify your shipping department immediately. Your shipping department may want to notify the carrier.

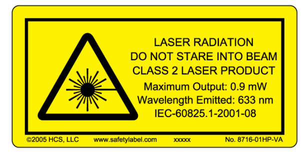
If the shipping carton is not damaged, carefully remove and identify all of the components listed below. Contact Bristol Instruments or your local representative if any of the components are missing. We recommend that you save the shipping carton for future storage or transportation.

The 621 system includes the following components:

- ✓ Wavelength meter head
- ✓ Three alignment feet assemblies (IR and MIR versions)
- ✓ Power cord (North America and Japan only)
- ✓ USB interface cable
- ✓ CD with operating software
- ✓ *User's Manual*

2 INITIAL INSTRUMENT SETUP

WARNING **This instrument is a Class 2 Laser Product. It contains a laser that emits radiation that can seriously damage your eyesight. Less than 15 µW of this laser light is emitted from the fiber-optic connector of the VIS and NIR versions, and less than 50 µW of this laser light is emitted from the aperture of the IR and MIR versions. Do not stare into this beam or view it directly with an optical instrument. The class 2 warning label, shown here, is located next to the fiber-optic input coupler (VIS and NIR) or the input aperture (IR and MIR) on the front panel of the model 621.**



General

1 Place the 621 Laser Wavelength Meter on a firm horizontal surface.

2 Make sure that there is at least 2 inches (50 mm) clearance on all sides of the instrument to allow for ventilation.

Power Supply Connections

WARNING **This is a Safety Class 1 product (provided with protective earth). The mains plug shall only be inserted in a socket outlet provided with a protective earth contact. Any interruption of the protective conductor inside or outside of the product is likely to make the product dangerous. Intentional interruption is prohibited.**

CAUTION Always use the three-prong AC power cord supplied with this instrument. Failure to ensure adequate earth grounding by not using this cord may cause instrument damage.

CAUTION *Do not* connect AC power until you have verified the line voltage is correct as described in step 1 below. Damage to the instrument could result.

CAUTION This instrument has autoranging line voltage input. Be sure the supply voltage is within the specified range.

1 Verify that the line power meets the requirements shown below.

- 90 to 260 VAC
- 47 to 63 Hz
- Protective Earth Ground

2 Connect the line-power cord to the power input connector on the instrument's back panel (Figure 2.1).

3 Connect the other end of the line-power cord to the power receptacle.

Figure 2.1: Back Panel of the Model 621

Signal Output Connections

The 621 Laser Wavelength Meter has the following connections for signal output and communications. These connections are located on the instrument's back panel (Figure 2.1).

- USB Port - interface to a PC for instrument control and data reporting.
- RJ-45 Connector - not used at this time.
- DB-9 Monitor Port - used to monitor the interference fringe pattern and instrument timing signals with an oscilloscope. A full description is provided in Appendix D.

1 Connect the USB interface cable to the USB port on the back panel of the instrument.

2 Connect the other end of the USB interface cable to a USB port on your PC.

Software Installation

The 621 Laser Wavelength Meter is controlled by, and data reported with, software that is provided on the *Wavelength Meter Application Software* CD. The following computer hardware is required to run the software and control the instrument.

- A PC running Microsoft Windows VISTA or Microsoft Windows XP
- A 1 GHz or higher microprocessor
- At least 128 MB of available RAM
- USB 1.1/2.0 port
- VGA monitor
- Mouse or other pointing device

Installation with Windows VISTA and Windows XP:

There are several embodiments and/or folder arrangements associated with Microsoft Windows. Some of these will vary, even on the same PC, if, for example, the theme is changed to Classic Windows. There are also varying degrees of automatically invoked installation Wizards that may skip over some steps, especially if the software is being reinstalled. An attempt has been made to describe these Microsoft Windows variations. However, since some of these variations are not covered, please contact Bristol Instruments at (585) 924-2620 or at service@bristol-inst.com if you have any questions during software installation.

1 Place the *Wavelength Meter Application Software* CD in the CD drive of your PC.

2 Open the CD file and observe the two folders ***Bristol Wavelength Meter*** and ***SiLabs***. Copy the ***Bristol Wavelength Meter*** folder and all its contents to a directory of your choice.

3 Open the CD file and open the folder ***SiLabs***. Double click on the file **CP210x_VCP_Win2K_XP_S2K3.exe**.

4 The InstallShield Wizard will be activated. Follow the Wizard's instructions until the installation is complete.

5 Confirm that the USB cable is connected to the 621 system and the PC as described above.

6 Power on the 621 Laser Wavelength Meter.

7 Navigate to the PC's **CONTROL PANEL**.

8 Navigate to the **DEVICE MANAGER**. The directory structure can vary depending on the version of Windows used or the theme chosen. Typically, navigation will go from **CONTROL PANEL** to **SYSTEM PROPERTIES** and then to the **HARDWARE** tab, which contains the **DEVICE MANAGER**. If you have trouble locating the **DEVICE MANAGER**, click on **Start** and select **Help**.

9 From within the **DEVICE MANAGER**, expand the **"Ports (COM and LPT)"** line by clicking on the small box to the left of the icon. The name of the driver will appear as **CP2101 USB to UART Bridge Controller (COMx)**, where **x** will be the COM port assigned to the driver. Remember this number for the next two steps.

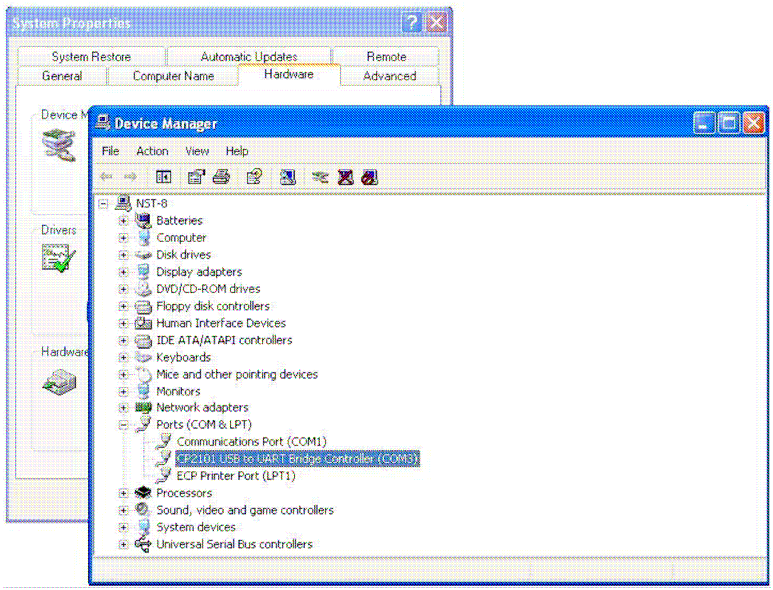


Figure 2.2: Device Manager in Windows XP

10 Navigate to the ***Bristol Wavelength Meter*** folder and open the ***WM621.ini*** file.

11 Under **[Com Port]** (the first entry), change the default number from 5 to the number recalled from Step 9, if different. **Do not change any other text in this file.**

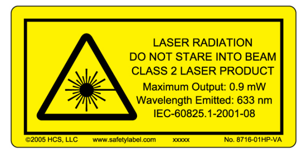
12 Save and close the ***WM621.ini*** file.

13 The software installation is now complete. Double click on the ***Wavelength Meter Application*** software in the applications folder and observe that the unit is displaying information on your PC.

3 LASER INPUT - VIS AND NIR VERSIONS

Fiber - Optic Laser Input

WARNING **This instrument is a Class 2 Laser Product. It contains a laser that emits radiation that can seriously damage your eyesight. Less than 15 μW of this laser light is emitted from the fiber-optic connector of the VIS and NIR versions, and less than 50 μW of this laser light is emitted from the aperture of the IR and MIR versions. Do not stare into this beam or view it directly with an optical instrument. The class 2 warning label, shown here, is located next to the fiber-optic input coupler (VIS and NIR) or the input aperture (IR and MIR) on the front panel of the model 621.**



WARNING ***Never*** **inspect or clean a fiber-optic cable without first disconnecting the entire cable assembly from the optical source. Failure to take this precaution can seriously damage your eyesight.**

CAUTION Use care in handling fiber-optic connectors. Always clean the fiber end prior to insertion into the instrument's fiber-optic connector for optimum performance. Failure to do so can result in damage to the instrument. To prolong instrument service life, it is advantageous to attach a new fiber-optic patchcord to the instrument and use its other end for the connect/disconnect procedure.

CAUTION Maximum safe input is 10 mW of optical power. Laser input power in excess of 10 mW can result in damage to the instrument.

The laser under test enters the 621 system through an FC/PC fiber-optic connector on the front panel of the instrument (Figure 3.1).

Figure 3.1: Front Panel of the VIS and NIR Versions of the Model 621

1 Ensure all fiber-optic connectors are clean and dry. To clean the end of fiber-optic connectors, first gently wipe the end with a lint-free swab dipped in isopropyl alcohol, then dry using clean compressed air.

I M P O R T A N T **Never clean a fiber-optic connector by other means, and be certain to clean the fiber-optic connector before each insertion. Damage not covered by the instrument's warranty will result from unclean or improperly cleaned fiber-optic connectors. A common practice to extend instrument fiber-optic connector life is to use a patch cord dedicated to the instrument, with its free-end used for the connect/disconnect procedure.**

2 Connect your fiber-optic cable to the pre-aligned FC/PC fiber-optic connector on the instrument's front panel. Make certain that the alignment key on the fiber-optic cable's connector (Figure 3.2) is properly seated in the slot of the input connector.

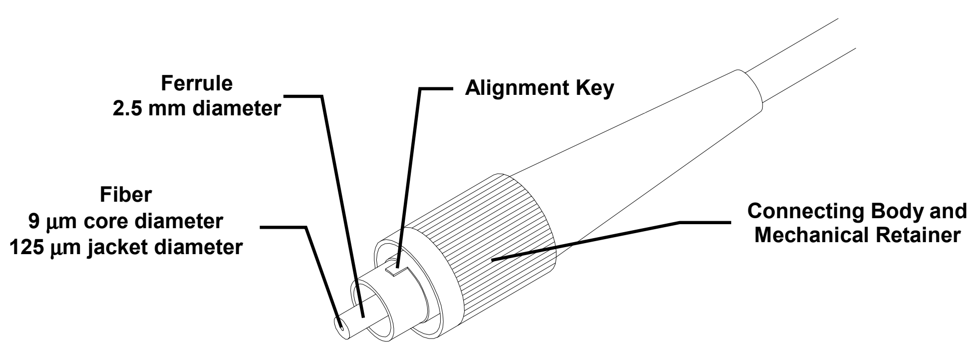
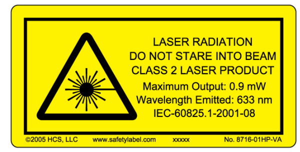


Figure 3.2: Basic Components of an FC-PC Fiber-Optic Connector

3 Tighten the mechanical retainer with a light to medium finger-tightness. Exceeding this torque may result in a poor connection or may damage the connector.

Free - Beam Laser Input

WARNING **This instrument is a Class 2 Laser Product. It contains a laser that emits radiation that can seriously damage your eyesight. Less than 15 µW of this laser light is emitted from the fiber-optic connector of the VIS and NIR versions, and less than 50 µW of this laser light is emitted from the aperture of the IR and MIR versions. Do not stare into this beam or view it directly with an optical instrument. The class 2 warning label, shown here, is located next to the fiber-optic input coupler (VIS and NIR) or the input aperture (IR and MIR) on the front panel of the model 621.**



WARNING ***Never* inspect or clean a fiber-optic cable without first disconnecting the entire cable assembly from the optical source. Failure to take this precaution can seriously damage your eyesight.**

CAUTION Use care in handling fiber-optic connectors. Always clean the fiber end prior to insertion into the instrument's fiber-optic connector for optimum performance. Failure to do so can result in damage to the instrument. To prolong instrument service life, it is advantageous to attach a new fiber-optic patchcord to the instrument and use its other end for the connect/disconnect procedure.

CAUTION Maximum safe input is 10 mW of optical power. Laser input power in excess of 10 mW can result in damage to the instrument.

If the laser under test has a free space beam, it must be launched into a fiber-optic patch cord for entry into the 621 Laser Wavelength Meter. The LC-1 Fiber-Optic Input Coupler offered by Bristol Instruments is a convenient method of doing this.

1 Install the LC-1 coupler in any two-axis (theta - phi) adjustable optical mount that can accommodate a 25.4 mm diameter optic. Optical mounts with X – Y, as well as theta – phi, adjustments may be convenient in some laboratory configurations.

2 Center the laser beam on the 2.5 mm input aperture of the LC-1 coupler. Using the angular adjustments of the optical mount, adjust the face of the beam coupler to be orthogonal to the laser beam

3 Two back reflections from the LC-1 coupler should be visible near the laser's output aperture. They can also be seen by placing a white card (with a hole in the center) in the beam path as shown in Figure 3.3. Adjust the optical mount until the reflections symmetrically straddle the input beam.

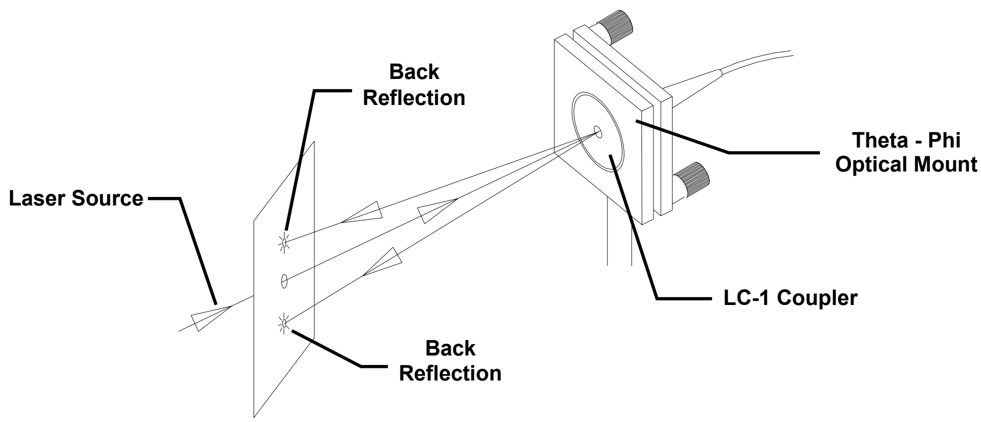


Figure 3.3: LC-1 Fiber-Optic Input Coupler Alignment

4 Maximize throughput by making minor angular adjustments with the optical mount and observing the bar-graph intensity meter of the model 621 Windows-based display.

4 LASER INPUT - IR AND MIR VERSIONS

Installation of Adjustment Feet

The IR and MIR versions of the 621 Laser Wavelength Meter is supplied with three leg/feet assemblies (Figure 4.1) that are used to precisely align the laser under test to the instrument's Michelson interferometer. Firmly attach the leg/feet assemblies to the 621 system by threading them onto the three 10-32 x ¼" threaded studs emerging from the underside of the chassis.

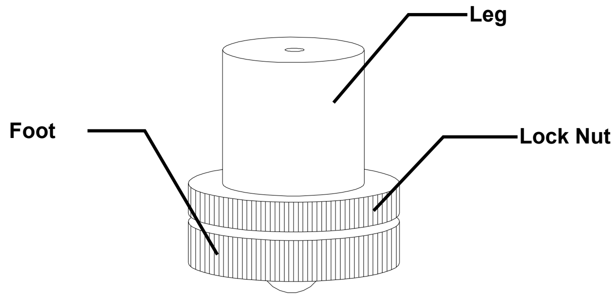


Figure 4.1: Model 621-IR/MIR Adjustment Feet

Alignment of Laser Under Test

WARNING **This instrument is a Class 2 Laser Product. It contains a laser that emits radiation that can seriously damage your eyesight. Less than 15 µW of this laser light is emitted from the fiber-optic connector of the VIS and NIR versions, and less than 50 µW of this laser light is emitted from the aperture of the IR and MIR versions. Do not stare into this beam or view it directly with an optical instrument. The class 2 warning label, shown here, is located next to the fiber-optic input coupler (VIS and NIR) or the input aperture (IR and MIR) on the front panel of the model 621.**

CAUTION Maximum safe input is 10 mW of optical power. Laser input power in excess of 10 mW can result in damage to the instrument.

The laser under test enters the IR and MIR versions of the 621 system through an input aperture on the front panel of the instrument (Figure 4.2).

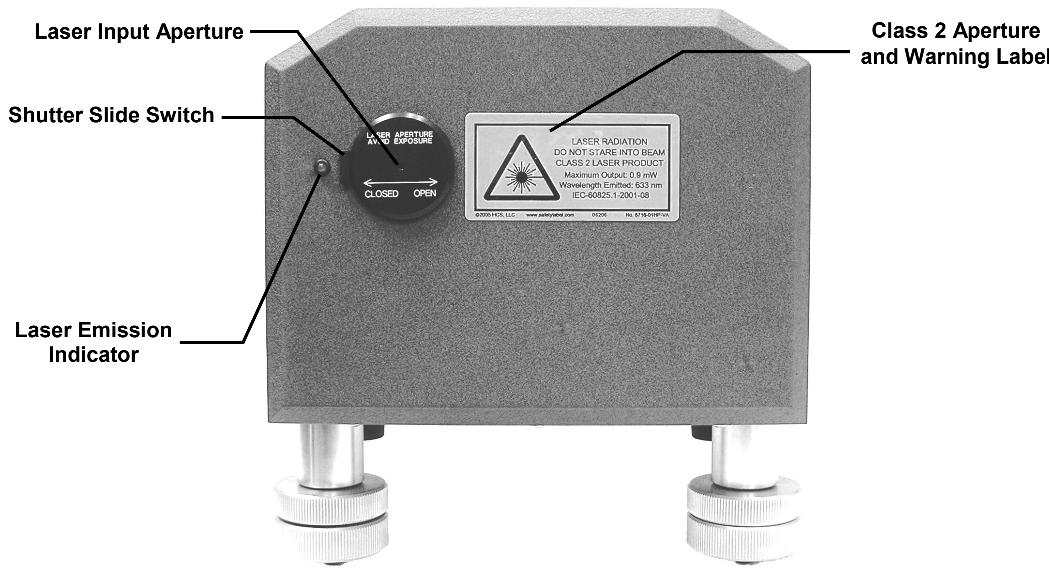


Figure 4.2: Front Panel of the IR and MIR versions of the Model 621

To facilitate alignment of the laser under test, the internal HeNe reference laser is emitted from the input aperture as a weak visible tracer beam. This tracer beam defines the optical path of the HeNe reference laser through the instrument's Michelson interferometer. In order to achieve the specified wavelength measurement accuracy, the laser under test must follow exactly the same path.

To block the red tracer beam from exiting the instrument, the input aperture is equipped with a manual shutter. The input aperture is open when the shutter slide switch is in the right-most position. The input aperture is closed when the shutter slide switch is in the left-most position.

WARNING **The power of the red tracer beam is less than 50 µW, but do not stare into the beam or view it directly with an optical instrument.**

1 Set the height of the three adjustment feet to mid-range.

2 With appropriate mirrors, adjust the laser under test to be parallel to the optical table at a height of approximately 5 ¾". Total travel of the 621 adjustment feet is ¾". It is not advisable to have the instrument too close to the top of the adjustment range.

3 Set the model 621 in front of the laser beam about 1 meter from the source.

4 Adjust the position and height of the two front adjustment feet so that the red tracer beam and the laser under test are superimposed at the input aperture.

5 Adjust the height and position of the back adjustment foot until the red tracer beam is directed back along the path of the laser under test.

IMPORTANT **In order to achieve the specified wavelength measurement accuracy, the red tracer beam and the laser under test must be precisely collinear (within 0.5 mm for the model 621A and within 1.5 mm for the model 621B) over a one meter path from the input aperture.**

6 Steps 4 and 5 may have to be repeated in order to optimize alignment.

IMPORTANT **Please note that the dynamic range of the IR and MIR versions of the model 621 is typically 10 to 15 times the minimum input level. A means of optical attenuation may be necessary to decrease the input power to a suitable level. Misaligning the 621 system is not an appropriate method of optical attenuation and will result in inaccurate wavelength measurements.**

7 Use the lock nuts on the adjustment feet to secure the model 621 when alignment is complete. This is done by rotating the lock nut upward to snugly contact the leg, thereby preventing the adjustment foot from rotating. The lock nut does not require substantial tightening.

5 USING THE WAVELENGTH METER

Operating Instructions

1 Turn on the power switch of the 621 Laser Wavelength Meter located on the back panel of the instrument.

2 Start the wavelength meter software application.

3 A Windows-based graphical user interface, shown in Figure 5.1, will display laser wavelength and power as soon as a suitable laser source is connected to the fiber-optic input.

NOTE Only the VIS and NIR versions of the 621 Laser Wavelength Meter measure laser power. The IR and MIR versions do not have the capability of measuring absolute laser power.

4 Use the PC's pointing device to activate the menus and buttons.

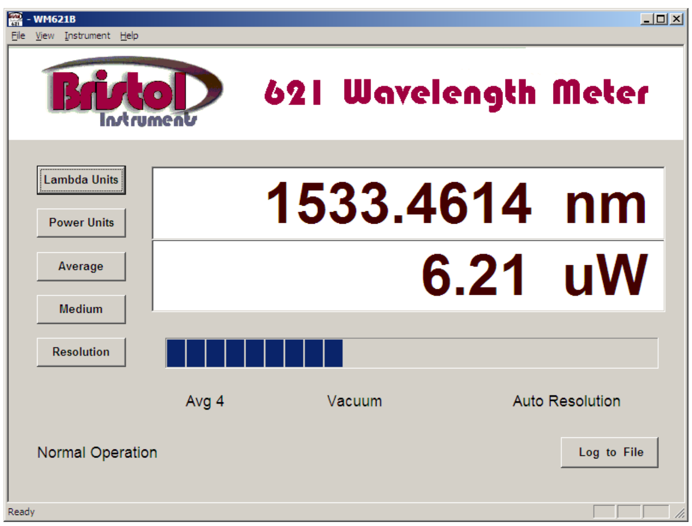


Figure 5.1: Model 621 Wavelength Meter Display (VIS and NIR versions)

Main Display

The 621 Laser Wavelength Meter display and controls are described in this section.

Data Display Reports the measured data as wavelength (nm), wavenumber (cm^{-1}), or frequency (GHz).

Reports the measured power in a linear scale (μW or mW) or in a logarithmic scale (dBm). Laser power is measured only by the VIS and NIR versions of the 621 system.

Intensity Meter This is a bar graph that displays the relative intensity of the laser input.

Lambda Units Cycles through the three choices of display units in succession; nanometers (nm), wavenumbers (cm^{-1}), or gigahertz (GHz).

Power Units Cycles through the two choices of display units in succession; either linear (μW or mW) or logarithmic (dBm). Laser power is measured only by the VIS and NIR versions of the 621 system.

Average Toggles between measurement data displayed as individual measurements (off) or as a running average (on). The number of measurements averaged is programmed using the *Wavelength Meter Settings* function described below.

The status of the *Average* function is reported below the *Data Display* (left).

Medium Toggles between measurement data displayed as vacuum or standard air (20°C, 760 mm Hg, and 0% relative humidity) values.

The status of the *Medium* function is reported below the *Data Display* (center).

Resolution Toggles between measurement data displayed with a resolution that is either determined automatically (Auto Mode), or set by the user (Fixed Mode).

The status of the *Resolution* function is reported below the *Data Display* (right).

In order to realize the specified accuracy from the 621A Laser Wavelength Meter, the bandwidth of the laser under test must be less than 1 GHz (FWHM). The bandwidth of the laser under test must be less than 10 GHz (FWHM) in order to achieve the specified accuracy from the model 621B. If the laser under test has a bandwidth that is larger than these values, the measurement accuracy will be reduced according to the following table.

Laser Bandwidth ($\Delta\nu$)	Accuracy
$\Delta\nu \leq 1$ GHz	± 0.2 ppm (± 0.0002 nm at 1000 nm)
1 GHz < $\Delta\nu \leq$ 10 GHz	± 1.0 ppm (± 0.001 nm at 1000 nm)
10 GHz < $\Delta\nu \leq$ 100 GHz	± 10 ppm (± 0.01 nm at 1000 nm)
100 GHz < $\Delta\nu \leq$ 1 THz	± 100 ppm (± 0.1 nm at 1000 nm)

Auto Mode: In this mode, the model 621 automatically determines the appropriate display resolution based on the bandwidth of the laser under test. Only significant digits are displayed.

Fixed Mode: In this mode, the display resolution is fixed by the user. The resolution value is programmed using the *Wavelength Meter Settings* function described below.

Although this mode allows the user to specify the number of digits to be displayed, the accuracy of the wavelength measurement is still limited by the bandwidth of the laser.

Log to File Enables the collection of wavelength data to a file with a *.csv format. When activated, this function asks for a file name in which wavelength data will be stored. After providing a file name, wavelength data is collected until the "Stop Log" function is activated. When data collection begins, the "Log to File" button changes to a "Stop Log" button.

View Menu

The menu choices under *View* are described below.

Settings This function allows the user to view and modify the instrument's various operating parameters. By selecting Settings under the *View Menu*, the following window will appear.

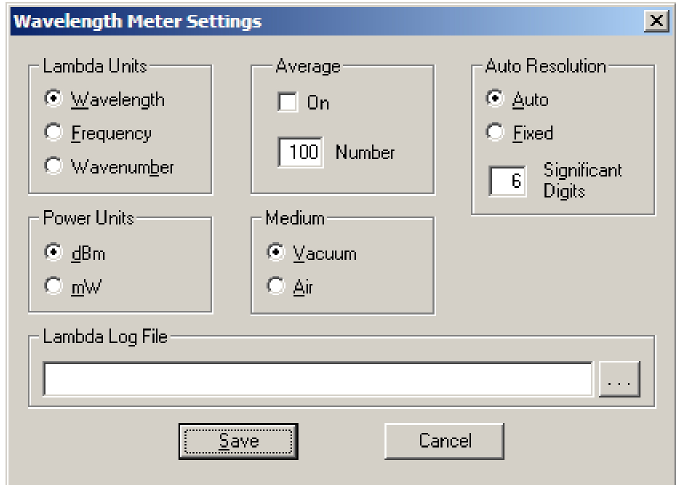


Figure 5.2: Wavelength Meter Settings Display (VIS and NIR versions)

Lambda Units: Wavelength (nm), frequency (GHz), or wavenumbers (cm^{-1}) can be selected.

Power Units: Either logarithmic (dBm) or linear (mW) units can be selected. Laser power is measured only by the VIS and NIR versions of the 621 system.

Average: The averaging function can be turned off or on (selected). When averaging is off, every measurement is displayed. When averaging is on, the instrument displays a running average.

The number of measurements averaged is set in the box adjacent to *Number*. A running average of as many as 100 measurements can be calculated.

Medium: Either vacuum or standard air (20°C, 760 mm Hg, and 0% relative humidity) can be selected.

Auto Resolution: The display resolution can be determined automatically (*Auto*) or set by the user (*Fixed*). In Fixed Mode, the number of digits displayed is set in the box adjacent to Sig*nificant Digits*.

Lambda Log File: Identifies a file name in which wavelength data will be stored when the *Log to File* function is activated.

6 REMOTE COMMUNICATIONS

LabVIEW Program Interface

Remote communication between a LabVIEW program and the 621 Laser Wavelength Meter is described in this section.

Two sample LabVIEW .vi's can be found in the Programming Interface folder on the *Wavelength Meter Application Software* CD. The sample in the LabVIEW Example I folder, getWavelength_v80.vi, uses calls to functions in the Windows dll CLDeviceIface.dll to access the wavelength meter. The functionality of this sample can be increased, or a new .vi can be written, by incorporating function calls using the following procedure.

1 Add a Library Function Node to the .vi.

2 Right click on this node to configure it.

3 In this configure window, click on Browse and find the path to the CLDevIface.dll. This file is found on the *Wavelength Meter Application Software* CD in the Bristol Wavelength Meter folder.

4 Click on the drop-down arrow next to Function Name to choose the function call needed. For this example, the CLOpenUSBSerialDevice call is used. This should be the first function call in the .vi.

5 Add the Return Type this function call expects. For the CLOpenUSBSerialDevice, the return type is long. Refer to the cldevdll.h file for the return types and parameters for each function. This file is found in the Bristol Wavelength Meter folder on the *Wavelength Meter Application Software* CD.

6 If the function call requires input parameters, these need to be specified by clicking Add a Parameter After. The parameter will be added in the parentheses.

7 When finished, the configuration window for the CLOpenUSBSerialDevice call should look like Figure 6.1.

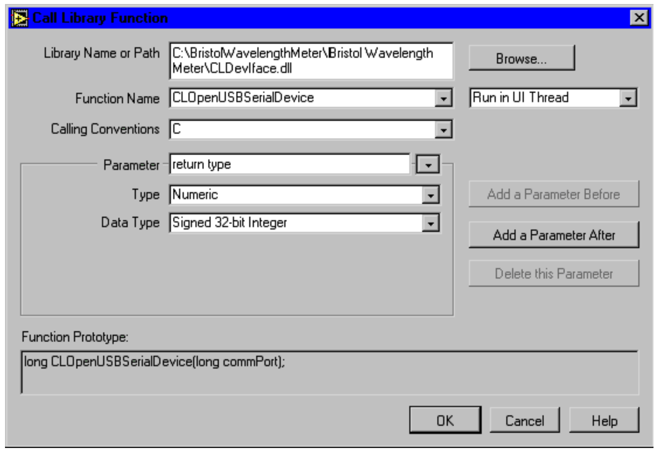


Figure 6.1: Configuration Window for the CLOpenUSBSerialDevice Call

8 Repeat the steps to configure any other function calls needed and then wire the nodes together. The sample .vi (getWavelength_v80.vi) shown in Figure 6.2 opens the serial device based on the com port specified, and then gets the wavelength every time the user clicks on "Get Wavelength".

9 When finished with the application, click on "Done" and the getWavelength_v80.vi will close the com port and exit. Note that the sample getWavelength_v60.vi does not close the com port and is only included here for reference.

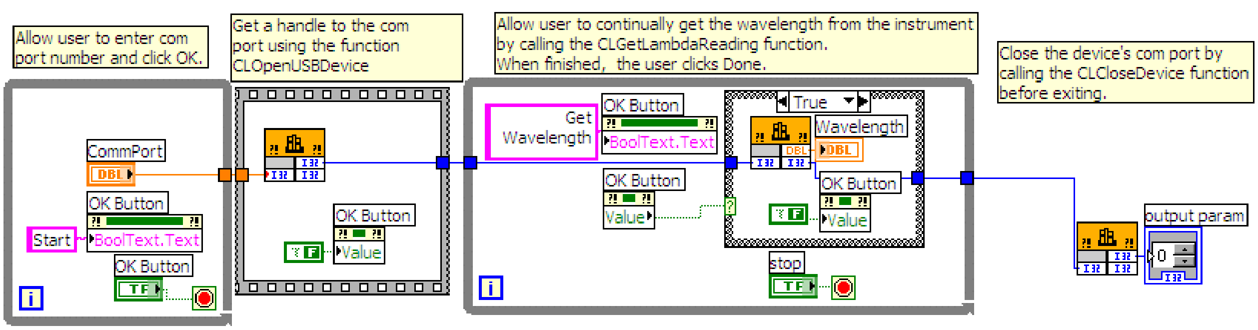


Figure 6.2: Sample getWavelength_v80.vi.

The sample .vi located in the LabVIEW Example II folder, BristolExample.vi, demonstrates how to interface LabVIEW directly to the 621 Laser Wavelength Meter without using the Windows dll. Because the communications functions that are contained in the dll are replicated in the .vi, the .vi is considerably more complex than the sample described above, which accesses the 621 Laser Wavelength Meter via the dll.

In order to interface directly with LabVIEW, the National Instruments' NI-VISA API/code library must be installed on the PC. This software is available for download at no charge from the National Instruments website.

Direct communication between LabVIEW and the 621 Laser Wavelength Meter can be initiated using the BristolExample.vi and the following instructions.

1 On the BristolExample.vi Front Panel, shown in Figure 6.3, choose the appropriate com port from the pull-down list under the label "Port." The appropriate com port is the same port that was indicated in the PC's device manager when the instrument software was initially installed.

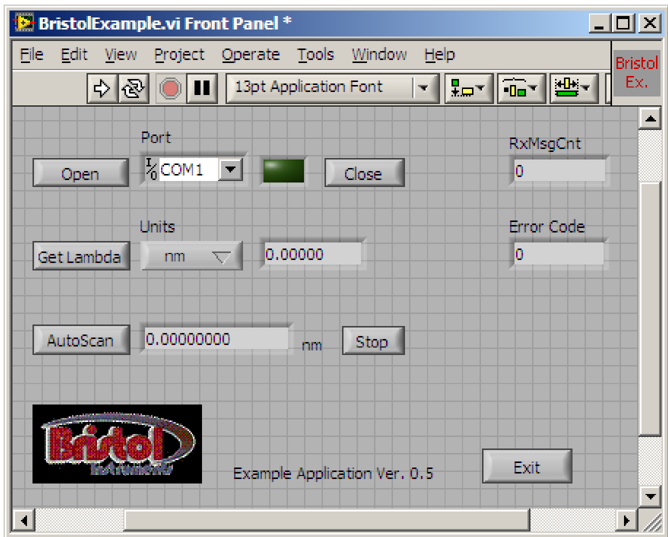


Figure 6.3: Sample BristolExample.vi.

2 If the appropriate com port is not shown, select "Refresh" at the end of the pull-down list. Also, make sure that the instrument is powered on and that the USB cable is properly connected.

3 Once the com port is selected, click on "Open." When the port is opened, the indicator to the right of the port list will change to a bright green.

There are two modes of communication available with the 621 Laser Wavelength Meter.

The Get Lambda mode consists of sending a request to the instrument and then waiting for a reply. Clicking on the "Get Lambda" button accesses this mode. The instrument returns the requested information in the units chosen on the "Units" pull-down menu.

Clicking on the "AutoScan" button accesses the second mode of communication. In this mode, the instrument automatically broadcasts every new wavelength measurement. The wavelength is given as the vacuum value in units of nanometers. To stop the broadcast of wavelength values, click on the "Stop" button.

When finished with the .vi, click on the "Stop" button if AutoScan is enabled. Then click on the "Exit" button to halt operation and properly close the com port.

Custom Program Interface - Windows dll Function Calls

Remote communication between a custom program and the 621 Laser Wavelength Meter, using the Windows dll "CLDevIface.dll" provided on the *Wavelength Meter Application Software* CD, is described in this section. For information describing the bypassing of the dll and communicating directly with the instrument from a custom program, see the next section: Custom Program Interface - Direct Communication.

The 621 Laser Wavelength Meter uses a Silicon Labs CP2101 USB to UART bridge controller. The drivers, provided on the *Wavelength Meter Application Software* CD, must be installed before remote communication can occur. This installation was done during the Software Installation described in Section 2. The device must first be opened using the CLOpenUSBSerialDevice command. This command returns a DeviceHandle that will be used for all other commands. The port must be closed at the end of your program by using a call to CLCloseDevice.

The following files will need to be examined and compiled with your custom program. These files are found on the *Wavelength Meter Application Software* CD in the Bristol Wavelength Meter folder.

\Bristol Wavelength Meter\CLDevIface.dll

and

\Bristol Wavelength Meter\Programming Interface\cldevdll.h

and

\Bristol Wavelength Meter\Programming Interface\Libraries\Microsoft_C\CLDevIFace.lib

or

\Bristol Wavelength Meter\Programming Interface\Libraries\Borland_C\CLDevIFace.lib

DECLSPEC int __cdecl CLOpenUSBSerialDevice(int ComNumber);

Function:

Opens the device using a USB Serial Port Interface.

Parameters:

ComNumber: the windows COM port number where the USB driver is installed.

Returns:

A valid CLDevice handle number, or –1 on failure. This device handle will be used with all commands to identify the port.

DECLSPEC int __cdecl CLCloseDevice(int DeviceHandle)

Function:

Closes the device using a USB Serial Port Interface. This command must be called at the end of the program to properly release the port.

Parameters:

DeviceHandle: the device handle returned by the call to the CLOpenUSBSerialDevice function.

Returns:

An integer error of –1 on failure, otherwise zero.

DECLSPEC int __cdecl CLSetMeasHBCallback(int DeviceHandle, MEASHBCALLBACK ProcessMeasHBData);

Function:

Set a user defined callback function to receive measurement information from the instrument when it is available.

Parameters:

DeviceHandle: a valid CLDevice handle.
ProcessMeasHBData: user supplied callback function.

Returns:

A valid CLDevice handle number, or –1 on failure.

DECLSPEC int __cdecl CLGet MeasurementData(int DevHandle, tsMeasurementDataType*data);

Function:

Set a user defined callback function to receive measurement information from the instrument when it is available.

Parameters:

DeviceHandle: a valid CLDevice handle.
Data: pointer to location to write data of type tsMeasurementDataType.

Returns:

–1 on failure.

DECLSPEC int __cdecl CLSetLambdaUnits(int DevHandle, unsigned int LambdaUnits);

Function:

Set the units returned from the CLGetLambdaReading.

Parameters:

DeviceHandle: a valid CLDevice handle.

LambdaUnits: 0 – nanometers (nm)
1 – gigahertz (GHz)
2 – inverse centimeters (cm^{-1})

Returns:

0

DECLSPEC int __cdecl CLSetPowerUnits(int DevHandle, unsigned int PowerUnits);

Function:

Set the units returned from the CLGetPowerReading.

Parameters:

DeviceHandle: a valid CLDevice handle.

PowerUnits: 0 – milliwatts (mw)
1 – decibels (dB)

Returns:

0

DECLSPEC double __cdecl CLGetLambdaReading(int DevHandle);

Function:

Get the current wavelength reading in units set by CLSetLambdaUnits.

Parameters:

DeviceHandle: a valid CLDevice handle.

Returns:

64 bit floating point wavelength reading in units set by CLSetLambdaUnits.

DECLSPEC float __cdecl CLGetPowerReading(int DevHandle);

Function:
Get the current power reading in units set by CLSetPowerUnits.

Parameters:
DeviceHandle: a valid CLDevice handle.

Returns:
32 bit floating point value of the power in units set by CLSetPowerUnits.

DECLSPEC int __cdecl CLSetMedium(int DevHandle, unsigned int medium);

Function:
Use to set the current medium for the CLGetLambdaReading function.

Parameters:
DeviceHandle: a valid CLDevice handle.
Medium: 0 – vacuum
1 – air

Returns:
0

Custom Program Interface - Direct Communication

Remote communication directly between a custom program and the 621 Laser Wavelength Meter, without using the Windows dll, is described in this section.

The 621 Laser Wavelength Meter uses a Silicon Labs CP2101 USB to UART bridge controller. The drivers, provided on the *Wavelength Meter Application Software* CD, must be installed before remote communication can occur. This installation was done during the Software Installation described in Section 2. The port must first be opened to access the instrument and then the port must be closed at the end of your program.

Acronyms used in this section:

APC	Application Personal Computer
ET	Escape Token
DSPB	DSP Board inside the wavelength meter
PC	Personal Computer
PPP	Point-to-Point Protocol
RFC	Request for Comment
ST	Start Token

Interface Overview

A scheme implemented within the PPP as defined in the document RFC 1549 will be utilized to uniquely identify the start of messages. RFC 1549 is a document created by the Network Working Group as the PPP was being established. The PPP uses a 0x7E to indicate the start/end of packets and 0x7D as an escape byte. The escape byte is used to indicate that the next byte has been converted to a different value using a well-known conversion function. This escaping of bytes allows the values of 0x7E and 0x7D to be sent as data but in a converted format. The receiver of the converted values must convert them back to their original values using the same well-known function. The APC-DSPB will utilize this same scheme.

In order to facilitate the detection of message boundaries, all messages will begin with the control byte 0x7E. Messages will not be terminated with a control byte. The 0x7E byte will be referred to as the start token, abbreviated “ST”. Since messages can contain the value 0x7E as part of its data, an escape byte, 0x7D, will be used to escape occurrences of 0x7E. The 0x7D will be used to escape itself in data as well. The 0x7D byte will be referred to as the escape token, abbreviated “ET”. Escaped values will be calculated as the exclusive OR of the byte with the value 0x20. This same operation is used to un-escape the byte.

0x7E XOR 0x20 = 0x5E	escaping an ST
0x5E XOR 0x20 = 0x7E	un-escaping an ST
0x7D XOR 0x20 = 0x5D	escaping an ET
0x5D XOR 0x20 = 0x7D	un-escaping an ET

The rules for escaping bytes on send are as follows:

Originating Data	Send Bytes on Interface As
Start of message	Send 0x7E
Non ST and non ET	Send byte value as is
0x7E as part of message data	Send 0x7D (0x20 XOR 0x7E)
0x7D as part of message data	Send 0x7D (0x20 XOR 0x7D)

The rules for un-escaping bytes on receive are as follows:

Received Bytes on Interface	Interpret Data As
ST	Start of message
Non ST and non ET	Receive byte value as is
ET	XOR of next byte with 0x20

Messages sent over the serial interface will begin with a message ID followed by a message sequence number, followed by a checksum, followed by the data length, followed by the message data.

Message sequence numbers will work through the values 0x01 to 0xFF and will rollover from 0xFF back to 0x01. The value 0x00 will never be used as a sequence number. The sender is responsible for initializing the sequence number to 0x01 for the first send, incrementing the sequence number by one for each successive send and rolling the sequence number over from 0xFF to 0x01. The receiver should flag an error if either 1) two successive messages are received that contain the same sequence number or 2) two successive messages are received that have a sequence number delta greater than one.

The checksum is the exclusive OR of all word (16 bit) values of the message without embedded escapes. For odd length messages, the final word is exclusive OR’d with the high byte set to zero. The checksum location is not included when calculating. When verifying, the entire message, included checksum, is used and should result in a checksum of zero.

Data lengths will indicate the length of the un-escaped data. The maximum data component length is 0xFF. This is a result of the maximum value that can be represented in a byte.

Communication Message Set

The message ID values are segmented as follows:

Range	Description
0x00 – 0x1F	Reserved
0x20 – 0x5F	APC to instrument message Ids
0x60 – 0x9F	Instrument to APC message Ids
0xA0 – 0xFF	Not used

The table below gives the message IDs for communication between the APC and the DSPB. All numeric values in this section are in hexadecimal format.

Message ID	Description	Comments
0x22	Set Power Units	0 = mW; 1 = dBm; see 0x40
0x23	Set Wavelength Units	0 = nm; 1 = GHz; 2 = cm-1; see 0x45
0x25	Set Medium	0 = vacuum; 1 = air; see 0x45
0x28	Set Auto Send	0 = off; 2 = measurement data
0x30	Read-back Measurement	Returns measurement data; see 0x60
0x40	Request Power	Returns power in current units; see 0x22
0x45	Read-back Lambda 1	Returns wavelength; uses current units and medium; see 0x23 and 0x25
0x60	Measurement Data Response	
0x61	Unit Info Response	
0x68	Power Read	
0x69	Read ADC Response	
0x6A	Scan Data Response	
0x6B	Lambda Response	
0x6C	Lambda 2 Response	
0x6E	Measurement heartbeat	
0x6F	Head best message	

Detailed examples

Note that byte offset and byte value fields are represented in hexadecimal values.

Set Wavelength Units

Direction: APC → DSPB

Byte Offset	Byte Name	Byte value(s)	Byte Description
00	Start Token	7E	Generic message start identifier
01	Message ID	23	Unique message identifier
02	Sequence Number	01 – FF	Sequential message identifier
3-4	Checksum	0000 – FFFF	Checksum (2 bytes)
5-6	Payload Length	04 00	Data length (2 bytes)
7-8	Pad	xxxx	Pad to word boundary (2 bytes)
9-C	Units		0 = nm; 1 = GHz; 2 = cm-1 (use unsigned int)

Request Wavelength (Read Back Lambda 1)

Direction: APC → DSPB

Byte Offset	Byte Name	Byte value(s)	Byte Description
00	Start Token	7E	Generic message start identifier
01	Message ID	45	Unique message identifier
02	Sequence Number	01 – FF	Sequential message identifier
3-4	Checksum	0000 – FFFF	Checksum (2 bytes)
5-6	Payload Length	0000	Data length (2 bytes)
7-8	Pad	xxxx	Pad to word boundary (2 bytes)

Response example: actual bytes sent and bytes received from wavelength meter using message 0x45, Read Back Lambda 1 (response wavelength = 0):

WRITE Length 9 bytes: 7E 45 06 40 07 00 00 05 01

READ Length 17 bytes: 7E 6B 35 63 B5 08 00 00 80 00 00 00 00 00 00 00 00

Request Power

Direction: APC → DSPB

Byte Offset	Byte Name	Byte value(s)	Byte Description
00	Start Token	7E	Generic message start identifier
01	Message ID	40	Unique message identifier
02	Sequence Number	01 – FF	Sequential message identifier
3-4	Checksum	0000 – FFFF	Checksum (2 bytes)
5-6	Payload Length	0000	Data length (2 bytes)
7-8	Pad	xxxx	Pad to word boundary (2 bytes)

Response example: actual bytes sent and bytes received from wavelength meter using message 0x40, Request Power:

WRITE Length 9 bytes: 7E 40 03 52 03 00 00 12 00

READ Length 15 bytes: 7E 68 84 2F B1 06 00 20 A8 86 01 C7 3E 20 A2

APPENDIX A1 – SPECIFICATIONS (VIS, NIR, IR)

		621A	621B
LASER TYPE		CW only	
WAVELENGTH			
Range		**VIS:** 350 - 1100 nm **NIR:** 500 - 1700 nm **IR:** 1.0 - 5.0 µm	
Absolute Accuracy		± 0.2 parts per million ± 0.0002 nm @ 1,000 nm ± 0.002 cm^{-1} @ 10,000 cm^{-1} ± 0.06 GHz @ 300,000 GHz	± 0.75 parts per million ± 0.00075 nm @ 1,000 nm ± 0.0075 cm^{-1} @ 10,000 cm^{-1} ± 0.225 GHz @ 300,000 GHz
Repeatability (1)	VIS/NIR IR	± 0.03 parts per million ± 0.06 parts per million	± 0.1 part per million
Calibration		Continuous with built-in stabilized single-frequency HeNe laser	Continuous with built-in standard HeNe laser
Display Resolution		9 digits	8 digits
Units		nm or cm^{-1} (vacuum or standard air), GHz	
POWER (VIS AND NIR)			
Absolute Accuracy (2)		± 15%	
Resolution		2%	
Units		mW, µW, dBm	
OPTICAL INPUT SIGNAL			
Maximum Laser Bandwidth (3)		1 GHz 0.003 nm at 1000 nm 0.03 cm^{-1}	10 GHz 0.03 nm at 1000 nm 0.3 cm^{-1}
Minimum Input (4)	VIS	1 mW at 350 nm, 20 µW at 650 nm, 75 µW at 1100 nm	
	NIR	300 µW at 500 nm, 10 µW at 1500 nm, 50 µW at 1700 nm	
	IR	750 µW at 1.0 µm, 100 µW at 3.0 µm, 1 mW at 5.0 µm	
Maximum Input (5)		10 mW	
MEASUREMENT RATE	VIS/NIR IR	4 Hz 2.5 Hz	10 Hz 2.5 Hz
OPTICAL INPUT	VIS/NIR IR	Pre-aligned FC/PC connector (9 µm core diameter) - optional free beam-to-fiber coupler Collimated beam, 2 mm diameter aperture, visible tracer beam exits aperture to facilitate alignment	
COMPUTER REQUIREMENTS		PC running Windows Vista or Windows XP with 1 GHz or higher microprocessor, at least 128 MB of available RAM, USB 1.1/2.0 port, VGA monitor, mouse or other pointing device	
INSTRUMENT INTERFACE		High-speed USB 2.0 interface with Windows-based display program Library of commands for custom and LabVIEW programming	
WARM-UP TIME		< 15 minutes	None
DIMENSIONS AND WEIGHT	VIS/NIR IR	6.5" W x 5.0" H x 15.0" L (165.1 mm x 127.0 mm x 381.0 mm) 14 lbs (6.4 kg) 6.5" W x 7.5" H x 15.0" L (165.1 mm x 190.5 mm x 381.0 mm) 14 lbs (6.4 kg)	
POWER REQUIREMENTS		90 to 260 VAC, 47 to 63 Hz	

1. Standard deviation for a 5 minute measurement period after the instrument has reached thermal equilibrium.
2. Calibration wavelength for VIS version is 633 nm. Calibration wavelength for NIR version is 1533 nm.
3. Bandwidth is FWHM. When bandwidth is greater, wavelength is automatically reported with reduced resolution.
4. Sensitivity at other wavelengths can be determined from graphs that are available upon request.
5. For IR version, the maximum input power level is about 10 to 15 times the minimum input power level.

APPENDIX A2 – SPECIFICATIONS (MIR)

		621B-MIR		
LASER TYPE		CW only		
WAVELENGTH				
	Range	4.0 – 11.0 μm		
	Absolute Accuracy	± 1.0 part per million ± 0.006 nm @ 6250 nm ± 0.002 cm^{-1} @ 1600 cm^{-1} ± 0.05 GHz @ 48000 GHz		
	Repeatability (1)	± 0.2 parts per million		
	Calibration	Continuous with built-in standard HeNe laser		
	Display Resolution	8 digits		
	Units	nm or cm^{-1} (vacuum or standard air), GHz		
OPTICAL INPUT SIGNAL				
	Maximum Laser Bandwidth (2)	10 GHz 1.3 nm at 6250 nm 0.3 cm^{-1}		
	Minimum Input (3)	0.4 mW at 4 μm	0.3 mW at 7.5 μm	0.5 mW at 11 μm
	Maximum Input	10 – 15 times minimum input level		
MEASUREMENT RATE		2.5 Hz		
OPTICAL INPUT		Collimated beam, 3 mm diameter aperture, visible tracer beam exits aperture to facilitate alignment		
COMPUTER REQUIREMENTS		PC running Windows Vista or Windows XP with 1 GHz or higher microprocessor, at least 128 MB of available RAM, USB 1.1/2.0 port, VGA monitor, mouse or other pointing device		
INSTRUMENT INTERFACE		High-speed USB 2.0 interface with Windows-based display program Library of commands for custom and LabVIEW programming		
WARM-UP TIME		None		
DIMENSIONS AND WEIGHT		6.5" W x 7.5" H x 15.0" L (165.1 mm x 190.5 mm x 381.0 mm)		14 lbs (6.4 kg)
POWER REQUIREMENTS		90 to 260 VAC, 47 to 63 Hz		

1. Standard deviation for a 5 minute measurement period after the instrument has reached thermal equilibrium.
2. Bandwidth is FWHM. When bandwidth is greater, wavelength is automatically reported with reduced resolution.
3. Sensitivity at other wavelengths can be determined from graphs that are available upon request.

APPENDIX B – WARRANTY & SERVICE

Warranty

The 621 Laser Wavelength Meter is warranted against defects in material and workmanship for a period of one year from date of shipment. During the warranty period, Bristol Instruments will repair, or at its option, replace parts that prove to be defective when the instrument is returned prepaid to Bristol Instruments. Before returning an instrument, always call Bristol Instruments for return authorization. The warranty will not apply if the instrument has been damaged by accident, misuse, or as a result of modification by persons other than Bristol Instruments personnel.

It is important to call Bristol Instruments or your local sales representative in advance of returning the unit, for a Return Authorization Number (RA#). This will ensure the prompt handling of the repair, as well as provide important tracking information.

The liability of Bristol Instruments, (except as to title) arising out of supplying said product, or its use, whether under the foregoing warranty, a claim of negligence, or otherwise, shall not in any case exceed the cost of correcting defects in the products as herein provided. Upon expiration of the warranty period specified herein, all liability shall terminate. The foregoing shall constitute the sole remedy of the buyer. In no event shall the seller be liable for consequential or special damages.

Service

There are no user serviceable parts inside the 621 Laser Wavelength Meter. All service and repair work for the instrument is to be done at Bristol Instruments.

If you have any questions about the operation of your 621 system or need to have your 621 system serviced, please call Bristol Instruments at (585) 924-2620. Or, you can contact us at service@bristol-inst.com.

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Victor, NY 14564
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APPENDIX C - TROUBLESHOOTING

The 621 Laser Wavelength Meter continually checks the quality of the signal it is analyzing and alerts the user to unusual conditions that might affect the precision of the wavelength measurement.

Error Messages

The following table provides a list of error messages and what they mean. If an error message is displayed, please call Bristol Instruments for additional information and/or instructions.

Error Message	Description
Wavelength Range Error	The wavelength (or frequency) measurement is outside of the specified range of the instrument.
Reference Laser Error	This indicates that the internal HeNe reference laser is not operating properly.
System Temperature Error	This indicates that the instrument temperature is outside of the allowed range of 0 to 50°C, or that there is a problem with the temperature sensor. Measurement will continue using the temperature at the end of the allowed range closest to the temperature sensor's reading.
Pressure Range Error	This indicates that the ambient pressure is outside of the allowed range of 500 to 900 mm Hg, or that there is a problem with the pressure sensor. Measurement will continue using the pressure at the end of the allowed range closest to the pressure sensor's reading.
Input Power Too High	The instrument has detected an input power level that is beyond the 10 mW allowed safe limit. Measurement will be suspended when this message is displayed.
Input Power Too Low	The instrument has detected an input power level below the minimum input required. Measurement will continue, but may not be within specified accuracy and/or repeatability limits.

APPENDIX D –MONITOR PORT

The Monitor Port on the rear panel of the model 621 provides a combination of analog and digital signals for observing the interference fringe signals and measurement timing signals using an oscilloscope. The Monitor Port uses a 9-pin, female, D-sub style connector shown in Figure D-1 and with a pinout described in the table below.

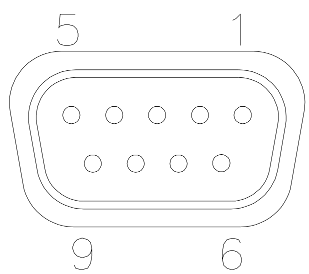

Figure D-1: Monitor Port

Pin Number	Signal Name	Signal Type	Description
1	REFERENCE	Analog	Reference laser sinusoidal interference fringe signal.
2	INPUT	Analog	Input laser sinusoidal interference fringe signal.
3			N/C
4			N/C
5	GROUND	-	Circuit ground.
6			N/C
7	TRIGGER	TTL	Scanning Window Trigger is a digital pulse with a duration that defines the maximum measurement interval for the interferometer scan. It goes active (H=active) during the allowed measurement interval.
8	ZOP	TTL	Zero Optical Path (ZOP) is a digital signal that indicates the location (timing) of equal path length in the arms of the interferometer. For scans of one direction, a positive edge indicates ZOP. For scans of the other direction, a negative edge indicates ZOP.
9	WINDOW	TTL	Scanning Window digital output is derived from the amplitude of the input laser fringe signal. When the p-p amplitude of this signal exceeds 1 volt, within the Trigger interval and before or spanning the ZOP transition, the Window is active (H=active). Depending on the character of the input laser fringes, there may be several Windows during a Trigger, but the first Window period to terminate after the ZOP transition is the only "Valid Window" used for the measurement.

EC and FCC
Declaration of Conformity

Manufacturer's Name: Bristol Instruments, Inc.

Manufacturer's Address: 7647 Main Street Fishers
Victor, NY 14564 USA

declares this product:

Product Name: Wavelength Meters

Model Number(s): 621A/B-VIS, 621A/B-NIR, 621A/B-IR, 621B-MIR

conforms to the following directives:

73/23/EEC
89/336/EMC

as a result of having been tested satisfactorily to the following standards:

Safety: EN-61010-1: IEC 1010-1

EMC: EN61326: 1997+A1: 1998+A2: 2001 **Class A**
EN61000-4-2: 1995 **±4KV Air & Contact**
EN61000-4-3: 2006 **3V/m 80MHz-2.7GHz**
EN61000-4-4: 1995 **±1KV**
EN61000-4-5: 1995 **±1KV(L-G) ±0.5KV(L-L)**
EN61000-4-6: 1996 **3V 0.15-80MHz**
EN61000-4-11: 1994 **1 Cycle 100%**
EN61000-3-2: 2000
EN61000-3-3: 2002

FCC: FCC per CFR 47 Part 15; Class A – Unintentional Radiators

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Date of Issue: January 9, 2009

Signed: ______________________________
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